

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed **project lifecycles** for automation and controls systems implementation, covering specification, design, and commissioning within a cGMP environment.
- Provided technical and **automation operational support** to Manufacturing and Facilities teams, resolving production issues and ensuring system availability.
- Implemented **software and hardware changes**, assessing and documenting the impact on manufacturing and facility operations.
- Supported the Validation Department by participating in initial design reviews, conducting **software testing**, and executing protocol reviews.
- Programmed **PLC and HMI applications** for automated process equipment, troubleshooting issues related to instrumentation and controls.
- Developed and maintained **Python scripts** for automating data extraction and report generation, processing approximately 1.5 TB of operational data weekly.
- Containerized Java Spring Boot applications using **Docker** and deployed them to an **AWS EC2** environment, improving deployment consistency.

SKILLS

Automation & Controls – PLC Programming, HMI Development, Instrumentation, Controls Systems, cGMP Environment, Commissioning, Start-up

Software Development – Java, Spring Boot, REST APIs, Python, SQL, Docker

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn

Operations & Support – Production Support, Incident Response, SLA Management, Change Control, Jira, Confluence, Splunk, Dynatrace

Project Management – Agile Methodologies, Scrum, Requirements Gathering, UAT Support, Technical Estimation

PROJECTS

Automated Batch Processing System Upgrade

- Engineered a new **PLC control system** for a critical batch processing line, incorporating enhanced safety interlocks and real-time monitoring capabilities.
- Developed **HMI interfaces** in parallel, providing operators with intuitive controls and detailed diagnostic information, reducing manual intervention time by **~20%**.
- Collaborated with the validation team to execute IQ/OQ protocols, ensuring the system met all cGMP requirements and compliance standards.

Python-based Production Data Analysis Pipeline

- Built a **Python ETL pipeline** using pandas and SQL to extract and transform data from disparate manufacturing systems, consolidating approximately 500 GB of daily production logs.
- Developed interactive **Power BI dashboards** to visualize key performance indicators, process deviations, and equipment uptime, enabling faster identification of operational bottlenecks for the manufacturing team.
- Implemented automated report generation scripts that delivered daily summaries to operations leads, reducing manual reporting effort by an estimated **8 hours per week**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)